

ST ATHAN, United Kingdom

WMO: 037160

Lat: 51.4053N

Lon: 3.4408W

Elev: 163

StdP: 14.61

Time Zone: 0.00 (EUW)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	28.0	30.5	21.3	16.1	31.5	23.8	18.1	33.2	56.2	48.2	50.7	48.7	9.1	70	0.635

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.1	75.3	64.2	71.8	62.6	69.0	61.4	65.8	72.2	64.2	68.9	63.0	67.1	10.0	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
63.6	88.5	67.7	62.3	84.6	66.2	61.2	81.2	65.1	30.6	72.4	29.4	69.1	28.5	67.3	79.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
34.2	27.5	23.5	DB	24.2	81.5	3.0	3.9	22.0	84.3	20.3	86.6	18.6	88.8	16.4	91.6	
			WB	22.9	69.6	2.9	3.3	20.8	72.0	19.1	73.9	17.4	75.8	15.3	78.2	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	51.6	42.5	42.0	44.8	49.0	53.8	59.2	62.1	61.6	58.6	53.5	47.5	43.5
	DBStd	8.64	5.61	5.12	4.70	4.51	4.35	3.83	3.97	3.15	3.73	4.82	5.40	6.03
	HDD50	1045	236	225	168	68	10	0	0	0	0	24	106	208
	HDD65	4942	699	643	626	479	348	181	109	114	194	358	525	666
	CDD50	1615	2	1	6	39	128	275	377	359	258	131	31	8
	CDD65	39	0	0	0	0	1	6	21	9	2	0	0	0
	CDH74	175	0	0	0	0	6	38	102	25	3	1	0	0
	CDH80	23	0	0	0	0	0	4	16	3	0	0	0	0
Wind	WSAvg	11.2	12.6	12.3	11.9	10.9	11.3	10.2	9.9	9.8	10.0	11.6	11.7	12.5
Precipitation	PrecAvg	37.3	3.6	2.5	2.8	2.2	2.4	2.4	2.8	3.2	3.5	3.8	4.1	3.9
	PrecMax	50.2	6.8	5.4	6.3	4.8	6.1	5.9	6.6	7.9	8.8	8.4	8.5	8.2
	PrecMin	27.2	0.4	0.0	0.5	0.4	0.2	0.2	0.5	0.9	1.0	0.4	1.5	1.2
	PrecStd	5.0	1.7	1.4	1.4	1.1	1.4	1.4	1.5	1.7	1.9	1.9	1.9	1.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	54.1	55.1	60.7	68.9	74.0	79.5	82.1	77.2	73.5	65.8	59.1	56.8
		MCWB	51.6	51.4	52.0	56.7	62.1	64.9	65.8	66.3	63.5	60.7	56.4	53.7
	2%	DB	52.4	52.2	56.0	63.1	68.3	72.8	76.7	71.8	69.4	62.9	57.5	54.6
		MCWB	50.6	49.3	49.8	54.0	59.5	62.4	65.2	64.1	61.7	58.8	55.2	52.5
	5%	DB	51.1	50.4	53.0	58.8	64.3	68.9	72.0	68.9	66.7	61.2	56.3	53.1
		MCWB	49.4	48.0	48.6	52.1	57.2	60.5	62.8	62.3	60.6	57.8	54.1	51.0
	10%	DB	49.8	49.1	51.1	55.8	61.0	65.9	68.9	66.9	64.7	59.9	54.9	51.6
		MCWB	48.1	47.1	47.8	50.4	55.2	59.0	61.6	61.2	59.7	56.8	52.7	49.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	52.6	52.2	52.9	57.9	63.7	68.0	68.4	68.0	65.6	62.1	57.5	54.5
		MCDB	53.3	54.1	57.5	66.0	71.5	76.6	78.5	74.3	69.7	64.8	58.4	55.7
	2%	WB	51.2	50.2	51.1	55.1	60.2	63.5	65.7	65.5	63.5	60.2	55.9	53.0
		MCDB	52.0	51.7	54.2	61.4	66.8	69.8	73.9	69.8	66.9	62.0	56.8	54.3
	5%	WB	49.8	48.9	49.8	53.0	58.1	61.5	64.0	64.0	62.1	58.9	54.6	51.7
		MCDB	50.7	50.1	52.1	57.7	63.1	67.2	70.3	66.9	65.4	60.6	55.8	52.8
	10%	WB	48.6	47.7	48.6	51.3	56.2	60.1	62.7	62.8	60.9	57.6	53.0	50.1
		MCDB	49.7	48.9	50.7	55.1	60.1	64.8	67.2	65.6	63.9	59.6	54.6	51.4
Mean Daily Temperature Range	5% DB	MDBR	7.9	8.8	10.3	12.0	11.8	11.3	11.1	10.7	11.4	9.6	8.6	8.2
		MCDBR	7.4	8.9	12.7	17.1	17.1	17.5	17.4	13.5	13.8	9.6	8.0	8.0
		MCWBR	7.5	7.2	8.7	10.2	10.3	9.6	8.8	8.0	8.3	7.2	7.3	8.0
	5% WB	MCDBR	7.6	8.1	10.8	14.2	14.8	14.6	14.5	11.1	11.1	8.4	7.5	7.8
		MCWBR	7.9	7.3	8.2	9.3	9.5	9.0	8.3	7.8	8.1	7.3	7.5	8.1
Clear-Sky Solar Irradiance	taub	0.330	0.344	0.375	0.392	0.398	0.388	0.387	0.374	0.366	0.354	0.328	0.320	
	taud	2.354	2.383	2.304	2.277	2.301	2.360	2.366	2.422	2.435	2.438	2.408	2.329	
	Ebn at Noon	214	246	259	268	272	275	273	272	260	239	214	196	
	Edh at Noon	20	26	33	38	39	37	36	33	29	24	19	18	
All-Sky Solar Radiation	RadAvg	249	470	833	1285	1577	1712	1604	1330	1002	567	301	194	
	RadStd	23	57	106	144	80	180	148	116	73	49	29	24	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (61 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page